

NARENDRANATH PANDA

R&D Architect

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Summary

Driving innovation and reliability in distributed systems, this **R&D Architect** brings **19+ years** of expertise in designing, operating, and scaling complex **Kubernetes platforms** across edge and cloud environments. Adept at leveraging **Kubernetes/EKS, AWS, and Infrastructure as Code (Terraform, Helm)**, the focus is on achieving operational excellence and delivering highly reliable, mission-critical systems. Expertise includes leading **incident management**, global oncall rotations, and **SLO/SLI-driven reliability programs**, successfully reducing downtime and deployment risk. Committed to building resilient platforms, enhancing developer velocity through **automation and GitOps**, and mentoring teams in SRE best practices within dynamic, product-centric organizations.

Work Experience

R&D Architect

July 2013 - Present

Nokia

- As a senior architect within a global R&D organization, I led strategic initiatives across platform modernization, SRE, security, and AIOps, delivering foundational improvements that enhanced reliability, accelerated development, and optimized costs.
- Orchestrated the zero-downtime migration for a distributed estate of over 1000(Edge Nodes) Kubernetes clusters, mitigating risks from hardware and software upgrades by engineering redundant services and robust failover mechanisms to ensure high availability for critical DevOps infrastructure.
- Led cross-functional teams to resolve fragmented modernization efforts, facilitating workshops and defining clear strategies (containerization, cloud readiness) to achieve stakeholder consensus and enable seamless transitions for **8-9** key projects. Developed **development processes** and **technical design** documentation.
- Architected and operated a reliable, highly available DC -Kubernetes platform (both onprem & AWS EKS) and **1000+** Edge k8s clusters using IAC (Terraform, Helm, GitOps), improving system resilience and resource efficiency. Designed the **technical architecture** and **data model** for scalability.
- Reduced deployment cycles by 50% and achieved 99.95% platform availability by architecting and implementing a standardized CI/CD(Gitlab-ci) workflow based on GitOps principles (ArgoCD), resolving slow, manual deployments across 8 projects.
- Streamlined product development by establishing and leading cross-team debugging "war rooms" and root cause analysis sessions, resolving critical blockers through permanent fixes and preventive automation. Enhanced **customer experience** by reducing **churn**.
- Directed a **15-engineer** DevOps team, implementing structured processes and DevOps best practices (IaC, CI/CD) to resolve inefficiencies and adopt a culture of proactive problem-solving and automation. Facilitated **Talent Acquisition** and **onboarding** for new team members.
- Designed and deployed a scalable observability and telemetry stack (Prometheus, Grafana, Loki, OpenTelemetry), integrating edge node telemetry and defining SLIs/SLOs to improve system reliability and reduce infra costs by **5%**. Implemented **monitoring tools** and **Sentry** for real-time issue detection.
- Created a POC on AI driven Git triage agent (RAG + Llama3) to automate issue categorization for over **10,000** items, projecting a **40-50%** reduction in manual effort and enabling **one-click** issue creation. Utilized **customer research** and **prototyping** to define requirements.
- Addressed escalating cloud costs by implementing a FinOps program that reduced spend by **15%** YoY and avoided a projected **\$500k** overspend through automated policies and provider negotiations. Focused on **build processes** and **release process** optimization.
- Automated and integrated compliance management for over **300** FOSS components, streamlining software releases and mitigating legal and security risks. Managed **inventory** and **codebase** integrity.
- Participated in a global **24/7** on-call rotation for mission-critical DevOps and platform services, troubleshooting, coordinating multi-team response, postmortems, and follow-up actions. Engaged in **change management** and **project management**.
- Drove **innovation** and efficiency across the **EMEA** region by implementing **Postgres** and **Redis** for enhanced data **modeling** and **scheduling**.

Sr. Developer

April 2011 - July 2013

Misys

- Co-engineered a new Java-based framework for the flagship LoanIQ product, replacing a legacy Smalltalk system and boosting system performance by 50% for key clients like Bank of America.
- Functioned as Scrum Master, optimizing bug triage and development workflows which led to a 70% reduction in defect resolution time.

IT Consultant

July 2006 - Mar 2011

Logica CMG

- Developed and delivered robust Java applications for diverse sectors, including education, oil & gas, and banking & finance (e.g., University & College Application System, GIT Air Liquide, LanaUt Swedbank).
- Wrote efficient, maintainable, and scalable Java code with best practices and coding standards

Education

Master of Technology -MTech WILP

2015 - 2017

Birla Institute of Technology and Science, Pilani

Bachelor of Engineering, Information Technology

2000 - 2004

Utkal University

Skills

Architecture

Cloud-Native Architecture | Microservices | Serverless | Platform Engineering | Solution Design | Scalable Architectures | Resilient Architectures | Distributed Systems | SaaS | Troubleshooting | Multi-tenant Platforms

DevOps & CI/CD

GitLab CI | GitHub Actions | ArgoCD | GitOps | CI/CD Pipelines | Infrastructure as Code | Automation | Ansible | Helm

Observability

Prometheus | Grafana | DataDog | Victoria Metrics | Open Telemetry | Loki | Thanos | AIOps | Kafka | FluentD | OpenSearch Database | Distributed Tracing

Tools and Technologies

Terraform | Vault | Jenkins | SonarQube | Harbor | Ceph | Rook | Jira | Confluence | MS Teams | Slack

Methodologies

Team Leadership | AIOps Strategy | DevSecOps | Cross-Functional Management | Agile/Scrum | Release Management

AI / ML / AIOps

AI Agent Design | Retrieval-Augmented Generation RAG | Large Language Models LLM Hosting & Prompting) | Model Context Protocol MCP | Vector Databases Pinecone/Chroma)

Language

English

Fluent

Achievements

Nokia Excellency Award

- Thanks for your contribution and mentorship 2025
- Successful NDAC DC migration 2025
- Excellence in Collaboration for MX Grid 2025

- Security supporter for SRE systems 2023
- Exceptional work for China setup DC 2023
- Recognized for exceptional work in setting up Data Centre in 2 months
- & many more

Logica Diamond Award

- Delivering software with quality